laws of quantum mechanics ensure that they are indistinguishable, not just in practice but in principle. They are perfect twins. If someone were to exchange the particles' positions (more precisely, exchange the two particles' probabilities of being located at any given position), there'd be absolutely no way to tell.

Thus, if we imagine starting with a particle located here,[40] and somehow put another particle of the same species into exactly the same quantum state (same probabilities for spin orientation, energy, and so on) at some distant location, the resulting particle would be indistinguishable from the original and the process would rightly be called quantum teleportation. Of course, were the original particle to survive the process intact, you might be tempted to call the process quantum cloning or, perhaps, quantum faxing. But as we'll see, the scientific realization of these ideas does not preserve the original particle—it is unavoidably modified during the teleportation process—so we won't be faced with this taxonomic dilemma.

A more pressing concern, and one that philosophers have considered closely in various forms, is whether what's true for an individual particle is true for an agglomeration. If you were able to teleport from one location to another every single particle that makes up your DeLorean, ensuring that the quantum state of each, including its relationship to all others, was reproduced with 100% fidelity, would you have succeeded in teleporting the vehicle? Although we have no empirical evidence to guide us, the theoretical case in support of having teleported the car is strong. Atomic and molecular arrangements determine how an object looks and feels, sounds and smells, and even tastes, so the resulting vehicle should be identical to the original DeLorean—bumps, nicks, squeaky left wing-door, musty smell from the family dog, all of it—and the car should take a sharp turn and respond to flooring the gas pedal exactly as the original did. The question of whether the vehicle actually is the original or, instead, is an exact duplicate, is of no concern. If you'd asked United Quantum Van Lines to ship your car by boat from New York to London but, unbeknownst to you, they teleported it in the manner described, you could never know the difference—even in principle.